



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2025	M.TECH	IIT Kharagpur	8.11 / 10
2018	M.Sc. Mathematics	Netaji Subhas Open University	61.90%
2016	B.Sc. Mathematics	University of Calcutta	48.62%
2013	A.I.S.S.C.E	CBSE	84.80%
2011	A.I.S.S.E	CBSE	9.4 / 10

INTERNSHIPS

Software Development Intern | ISRO (Space Applications Centre), Ahmedabad [May '24 - July '24]
To develop an interactive Graphical User Interface for Surface Water and Ocean Topography Level 2 and Level 3 data for the year 2023.
Tkinter | cartopy | xarray | geopandas | tqdm | concurrent.futures | matplotlib | pandas | numpy | Spyder | Linux
• The user is required to input the **min, max latitude and longitude** values as well as the **start and end dates** in the designated fields.
• Next the user needs to select the desired parameters from the **drop-down menu** and designate the folder path for the loading of data.
• The **netCDF** files that match the user-specified criteria will be loaded and the process will be tracked real-time using a **progressbar**.
• The GUI allows to select and **plot multiple files** from the valid list and provides the option to **save selected files as text** with a single click.

PROJECTS

Real-time weather query application | GenerativeAI | MTP | Prof. M.K. Dash [Aug'24 - Present]
• Designing a speech-to-text application that generates real-time weather response about a specific location based on the user's query.
• Integrating API keys from multiple data sources, enabling the retrieval of accurate and up-to-date weather information.
• Integrating advanced speech recognition techniques to facilitate smooth voice based query handling and enhance user interaction.

Potato Disease Classification | CNN | Deep Learning [July '24 - Present]
TensorFlow | Keras | Matplotlib | Numpy | FastAPI | tf serving | React JS | GCP | PyCharm | Jupyter Notebook
• Utilized **tf.data.Dataset** to reshuffle, batch (**32 elements**); applied resize, rescale and data augmentation within the **Sequential Model**.
• Added a series of **Conv2D** and **MaxPooling layers**, each using **activation function = 'relu'**, to extract features and reduce dimensionality.
• The model output is flattened and passed through dense layers with **activation = 'softmax'** for classification, **number of classes = 3**.
• Compiled the Convolutional Neural Network model, by '**adam**' optimizer and **Sparse Categorical Entropy loss** and evaluation metric.
• Results with **25 epochs** : yielding accuracy of **0.95**, loss of **0.15** while the validation data showed an accuracy of **0.93** and loss of **0.18**.

Resume Tracking System | Generative AI [March '24 - April '24]
Gemini API | Gemini-Pro | google-generativeai | PyPDF2 | Streamlit | Visual Studio Code
• Integrated **Google Generative AI** to evaluate resumes against job roles, delivering a detailed percentage match with profile summary.
• Utilized PyPDF2 for extracting text from resumes and crafted a prompt to facilitate precise resume evaluation with gemini-pro model.

SMS Spam Classification | Machine Learning | Natural Language Processing [Dec '23 - Jan '24]
Numpy | Pandas | Seaborn | Matplotlib | NLTK | Wordcloud | sklearn | Streamlit | PyCharm | Jupyter Notebook
• Performed Exploratory Data Analysis, text preprocessing, model building and evaluation and converted to a website using Streamlit.
• Implemented Gaussian, Multinomial and Bernoulli Naive Bayes algorithms; evaluated using Accuracy, Precision and Confusion matrix.
• Used **Bag-of-Words** and **Tf-idf vectorizer**; **Tf-idf** with **Multinomial Naive Bayes** yielded precision score of **1.0** and accuracy of **0.97**.

WORK EXPERIENCES

Mathematics lecturer | Narula Institute of Technology (JIS Group), Agarpara, Kolkata, West Bengal [Oct '21 - Jun '23]
• Instructed the Engineering Mathematics curriculum encompassing **Linear Algebra, Calculus, Probability & Statistics** to B.Tech students.

SKILLS AND EXPERTISE

Language : Python | **Technical Knowledge** : Machine Learning, NLP, Deep Learning, Generative AI, SQL, HTML, CSS, Javascript | **Operating Systems and Software** : Windows, Linux, MS Office, Power BI | **Soft Skills**: Leadership, Networking, Problem solving, Communication

COURSEWORK INFORMATION

Data Analytics | Computational Methods for Earth System Science | Simulation Lab | Satellite Remote Sensing | Geophysical Fluid Dynamics | Weather Analysis and Prediction | Coastal Processes | Global Climate system and Cloud Precipitation Processes

CERTIFICATIONS

LangChain: Chat with your Data | *DeepLearning.AI* : Document loading & splitting, vector stores & embeddings, RAG.
Data Analysis with Python | *IBM Skills Network* | *Coursera*: Exploratory Data Analysis, Model Development, Evaluation and Refinement.
Machine Learning with Python | *IBM Skills Network* | *Coursera* : Basics of machine learning, key algorithms and evaluation metrics.
Data Analysis with MS Excel | *Coursera Project Network* : Data manipulation, visualization and application of statistical functions.

POSITIONS OF RESPONSIBILITY

Student Head | Founding Member | Training and Placement Cell, CORAL | IIT Kharagpur [Nov '23 - Present]
• Represented department at **Academia-Industry Conclave** (Dec '23, July '24), organized by **Career Development Centre, IIT Kharagpur**.
• Worked closely with esteemed professors (CDC Coordinators, CORAL), engaging with potential employers from diverse backgrounds, fostering connections between Earth System Science domain and its growing industry with a focus on AI, ML and Generative AI.
• Oversaw and managed the official *LinkedIn Page* of the department, significantly enhancing its online presence and boosting engagement.